Chapter

9

TRANSMISSION OF POWER

SKNDKA

Learning Objectives

1. Students to know about the Transmission of power.
2. To know about the types of transmission of power like open belt and cross belt.
3. To know about the gear transmission and their types

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9.1 Introduction

- A source of power is always needed in workshop processes particularly in cutting and forming of metals. Electricity as a means of conveying power to machinery is widely adopted. The electrical energy is converted into rotational energy by means of an electric motor and the machine converts the input of rotational energy into various forms useful work.
- It is the movement of energy from its place of generation to a location where it is applied to perform useful work.

9.2 Power Transmission

- Power transmission devices are very commonly used to transmit power from one shaft to another. Belt, chains and gears are used for this purpose. When the distance between the shafts is large, belts and ropes are used and for intermediate distance chains can be used. Gear drive is used for short distances.
- When power is transmitted by gears and chain , there is no slip in velocity ratio. It is called positive drive. When power is transmitted by a belt drive, there is always a possibility of some slip between the belt and the faces of the pulleys, so the character of motion transmitted is non-positive.

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